

App Dev Project Report

1. Student Details

Name: Debanjan Sahoo

Roll Number: 23f2000223

Email: 23f2000223@ds.study.iitm.ac.in

About Me:

I am a BS Degree student at IIT Madras with a strong interest in backend development, database design, and data-driven web applications. I enjoy building structured, scalable systems that solve real-world institutional problems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement

Institutes require an efficient digital system to manage campus recruitment activities involving **Admin (Institute Placement Cell), Companies, and Students.**

Manual systems using spreadsheets and emails lead to:

- Difficulty in approving companies and drives
- Duplicate student applications
- Poor tracking of application status

The objective was to design and build a **role-based web application** that manages the complete placement lifecycle.

Approach

The application was built using a **Flask-based modular architecture** with SQLite database integration.

Key implementation steps:

1. Designed normalized relational database schema using SQLAlchemy.
2. Implemented role-based authentication (Admin, Company, Student).
3. Created separate dashboards for each role.

The database is created programmatically using SQLAlchemy models (no manual DB creation)

3. AI / LLM Declaration

I used **ChatGPT (GPT-5)** for:

- Improving HTML structure and formatting consistency
- Debugging Flask routing errors
- Suggesting improvements in SQLAlchemy relationship definitions
- Improving variable naming and documentation clarity

Extent of AI usage: **Approximately 15–20%**, limited to assistance and code suggestions.

All final implementation logic, debugging, database relationships, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for SQLite database
SQLite	Relational database
Jinja2	HTML templating
Bootstrap 5	UI styling and responsiveness
Flask-Login	Authentication and session management
WTForms	Form validation
Werkzeug Security	Password hashing
Python	Core programming language

5. Database Schema / ER Diagram

Tables Implemented

1. User

- id
- name
- email ,password

2. Student Profile

- id
- user_id
- roll_number
- branch
- resume_path

3. Company Profile

- id
- user_id
- company_name
- hr_contact
- website
- approval_status (Pending / Approved / Rejected)

4. Placement Drive

- id
- company_id
- job_title
- job_description
- eligibility
- deadline
- status (Pending / Approved / Closed)

5. Application

- id
- student_id
- drive_id
- application_date
- status (Applied / Shortlisted / Selected / Rejected)

Relationships

- One-to-One → User ↔ StudentProfile
- One-to-One → User ↔ CompanyProfile
- One-to-Many → CompanyProfile → PlacementDrive
- One-to-Many → StudentProfile → Application
- One-to-Many → PlacementDrive → Application



6. API Resource Endpoints

Endpoint	Method	Description
/login	GET/POST	User login
/register/student	GET/POST	Student registration
/register/company	GET/POST	Company registration
/admin/dashboard	GET	Admin dashboard
/admin/approve_company/	POST	Approve company
/admin/approve_drive/	POST	Approve placement drive
/company/dashboard	GET	Company dashboard
/company/create_drive	GET/POST	Create drive
/student/dashboard	GET	Student dashboard
/student/apply/<drive_id>	POST	Apply for drive

7. Drive Link of Presentation Video

Google Drive Link:

https://drive.google.com/file/d/1VoR7qQfLywSW1LQUkOVk_ZS156OPPLgT/view?usp=sharing